QuantFusion

AI Quant Firm — Complete Blueprint

A one-person AI hedge fund that replaces every role in a traditional quant firm with autonomous AI agents. Built on Palantir's connected intelligence and Anduril's autonomous execution.

Architecture	Palantir Ontology (connected data) + Anduril Lattice (autonomous execution)
Strategy	Multi-timeframe mean reversion scalping on crypto futures
Markets	BTC, ETH, SOL, XRP on MEXC (primary) and Bybit (backup)
AI Agents	7 autonomous agents + 1 human CEO
Target	0.5% daily (scaling to 1%) through compounding
Risk Model	6-layer risk cage with kill switch

1. The Vision

QuantFusion operates as a fully autonomous AI-powered quant trading firm. The core thesis is simple: rather than hiring people, we hire AI. Every role in a traditional hedge fund — researcher, analyst, portfolio manager, risk manager, trader, data engineer, performance analyst — is replaced by a specialised AI agent that works 24/7, never sleeps, never gets emotional, and never takes holidays.

The human (CEO) does one thing: set the strategy, define risk limits, and review weekly performance. Everything else is autonomous. The AI employees work for the cost of API calls and electricity.

The Palantir Foundation

Palantir's AIP platform is built on the concept of an Ontology — a unified data layer where everything is connected to everything else. A soldier's position is linked to nearby threats, available weapons, rules of engagement, and mission objectives. Nothing exists in isolation.

In QuantFusion, the Ontology works identically. The BTC price is connected to the RSI reading, which is connected to the MACD histogram, which is connected to the volume spike, which is connected to the funding rate, which is connected to the news sentiment, which is connected to the current session time, which is connected to existing open positions, which is connected to the daily P&L; and drawdown level. One giant interconnected web.

When any single piece of data changes, the entire web updates. A negative news article doesn't just update the Research Agent — it ripples through the entire system. The Risk Agent tightens limits. The Position Agent reduces size. The Math Agent adjusts confidence. Every agent reacts because everything is connected.

The Anduril Foundation

Anduril's Lattice platform enables autonomous execution at machine speed with intent-based commands. A commander doesn't micro-manage each drone — they say 'protect this airbase' and Lattice figures out how: which drones to deploy, where to position them, how to coordinate, when to escalate.

In QuantFusion, the CEO doesn't say 'buy 0.5 BTC at $64,200 with a stop at $63,400.' The CEO says 'trade BTC mean reversion, 0.5% daily target, 3% max daily loss' and the system handles everything autonomously — when to enter, how much, where to stop, when to add DCA layers, when to take profit, when to stop for the day.

Lattice also provides mesh resilience — no single point of failure. If one drone goes down, others cover. In QuantFusion, if MEXC's API dies, the system reroutes to Bybit. If a news feed dies, the system trades on math alone but raises its confirmation threshold. Every position has exchange-side stop losses that survive even a complete server crash.

2. Firm Structure — Traditional vs AI

Every role in a traditional quant firm maps directly to an AI agent in QuantFusion. The hierarchy, chain of command, and information flow remain identical — only the employees change from human to AI.

Traditional Role	AI Agent	Status	Core Responsibility
CEO / CIO	You (Ibrahim)	Human	Sets strategy, risk appetite, reviews performance
Head of Research	Research Agent	Autonomous	Reads news, sentiment scoring, confidence multiplier
Quant Analyst	Math Agent	Autonomous	Calculates indicators, runs checklist, produces score
Portfolio Manager	Position Agent	Autonomous	Position sizing, DCA layers, portfolio exposure
Risk Manager	Risk Agent	Autonomous	Enforces all limits, absolute veto power, kill switch
Trader	Execution Agent	Autonomous	Places orders via exchange API, manages stops/TPs
Data Engineer	Data Agent	Autonomous	Real-time data feeds, WebSocket, backup sources
Performance Analyst	Analytics Agent	Autonomous	Logs trades, tracks metrics, feeds learnings back

Trade flow — how a decision moves through the firm

Every trade passes through this exact chain. No shortcuts. No agent can skip a step. Information flows right, decisions flow right, results feed back left.

3. Agent Roles — Detailed Specifications

3.1 Data Agent — The Plumbing

The Data Agent is the foundation of the entire system. Without clean, real-time data, nothing else works. It maintains persistent connections to exchange APIs and aggregates all incoming information into the shared Ontology.

Data Source	Method	Update Frequency	Purpose
Price (BTC/ETH/SOL/XRP)	WebSocket stream	Real-time (every tick)	Core price data for all calculations
Volume	WebSocket stream	Per candle close	Volume ratio calculation (vs MA20)
Order book depth	REST API polling	Every 5 seconds	Liquidity analysis, slippage estimation
Funding rates	REST API polling	Every 8 hours	Crowd positioning (long vs short)
Open interest	REST API polling	Every 1 minute	Market participation tracking
News feeds	API + scraping	Every 5 minutes	Sentiment analysis input
On-chain data	API (future phase)	Every 15 minutes	Whale movement detection

Mesh resilience: If MEXC WebSocket drops, switch to Bybit within 2 seconds. If both drop, flag stale data and pause trading. Never trade on old data.

3.2 Math Agent — The Brain

The Math Agent is the quantitative core. It receives raw data from the Ontology, calculates all technical indicators across all timeframes, and produces a raw score for every potential trade setup.

Indicator	Settings	Role in System
RSI (dual)	RSI 6 (fast) + RSI 24 (slow)	Primary trigger — wakes up the system
MACD	12, 26, 9 (default)	Momentum confirmation — are sellers/buyers fading?
EMA 200	200-period exponential MA	Trend filter — bull or bear market?
Bollinger Bands	20-period, 2 std dev	Volatility confirmation — price at the wall?
ATR	14-period	Stop loss sizing — 1.5x ATR from entry
Volume ratio	Current vs MA(20)	Conviction check — crowd behind the move?

The Math Agent produces a raw score (0-5) from the confirmation checklist. It has no opinions — only numbers. The score is then passed to the Research Agent for contextual adjustment.

3.3 Research Agent — The Context Layer

The Research Agent is the LLM-powered intelligence layer. While the Math Agent sees numbers, the Research Agent reads and understands context. It's the difference between 'RSI is 14' and 'RSI is 14 because the SEC just sued the largest exchange.'

It takes the Math Agent's raw score and applies a confidence multiplier based on news context. This is Bayesian updating: prior belief (math score) multiplied by new evidence (news) equals posterior belief (adjusted score).

Scenario	Math Score	News Context	Multiplier	Adjusted
Normal dip	4/5	Profit-taking, ETF inflows strong	1.1x	4.4 = STRONG
Uncertain	4/5	Mixed signals, regulatory noise	0.8x	3.2 = MODERATE
Crisis	4/5	SEC sues exchange, systemic risk	0.15x	0.6 = NO TRADE

3.4 Position Agent — The Portfolio Manager

The Position Agent decides HOW MUCH to trade. It never decides what or when — that comes from Math and Research. It takes the approved signal and calculates the optimal position size, manages DCA layers, and considers the portfolio as a whole.

Position sizing formula: risk_amount = account_balance x 0.25% (per-trade cap). stop_distance = 1.5 x ATR. position_size = risk_amount / stop_distance.

DCA management: Layer 1 at RSI threshold (1/3 position). Layer 2 if RSI drops 5 more points (add 1/3). Layer 3 at extreme oversold (final 1/3). Each layer passes its own checklist independently.

Portfolio awareness: Checks existing positions and exposure. If already 2/3 of risk budget is deployed, the third position gets reduced size. Never overcommits.

3.5 Risk Agent — The Guardian

The Risk Agent is the most important agent in the system because its job is keeping the firm alive. It has absolute veto power — it can override every other agent and shut down all trading instantly. It is the only agent whose decisions cannot be overridden.

Rule	Limit	What happens when triggered
Per-trade risk cap	0.25% of account	Position sized down — never exceeded
Max concurrent positions	3 trades open	4th trade blocked regardless of score
Correlation check	> 0.85 blocked	Can't long BTC + ETH simultaneously
Losing streak breaker	3 consecutive losses	Pause trading for 30 minutes
Session streak breaker	5 consecutive losses	Stop trading for the entire session
Daily loss limit	-1.5% of account	ALL trading stops until tomorrow
Daily profit target	+0.5% (scaling to 1%)	ALL trading stops — lock the win
Max drawdown kill switch	-8% from account peak	ALL trading stops. CEO must review and restart.

Session thresholds	Varies by session	Asia: 5/5 needed. London: 4/5. Prime: 3/5.

3.6 Execution Agent — The Trader

The Execution Agent is the hands of the firm. It doesn't think, analyse, or decide — it executes what the other agents have approved. It connects to exchange APIs and places orders with precision.

Function	Details
Order placement	Limit orders at specified entry price. Never market orders (avoid slippage).
Stop loss	Set ON THE EXCHANGE — survives server crash. Always 1.5x ATR from entry.
Take profit	Set when RSI target is reached (45-50 zone = mean reversion complete).
DCA management	Monitors for layer 2/3 entries. Adjusts stops on all layers when adding.
Order monitoring	Tracks fill status. If limit order not filled in 5 minutes, cancel and reassess.
Exchange failover	If MEXC API returns errors 3x, switch to Bybit. Anduril mesh principle.

3.7 Analytics Agent — The Learning Loop

The Analytics Agent closes the feedback loop. After every trade, it logs every detail and calculates performance metrics. Over time, it learns which indicators are most predictive and adjusts weights. This is the Palantir Ontology feedback: decisions lead to outcomes, outcomes lead to learning, learning leads to better decisions.

Metric	What it tracks	Target
Win rate	Winning trades / total trades	Above 60%
Expectancy	(win_rate x avg_win) - (loss_rate x avg_loss)	Positive always
Sharpe ratio	Return / risk (annualised)	Above 2.0
Profit factor	Total profit / total loss	Above 1.5
Best session	Which session produces highest returns	Optimise allocation
Best indicator	Which confirmation is most predictive	Adjust weights
Avg hold time	How long positions stay open	Optimise exits
Max drawdown	Worst peak-to-trough decline	Below -8%

4. Strategy — Multi-Timeframe Mean Reversion

The system watches three timeframes simultaneously and catches mean reversion opportunities at different scales. Higher timeframes always override lower ones.

Timeframe	Role	RSI Trigger	Hold Time	Trades/Day	Profit/Trade
5-minute	Sniper scalps	< 25 or > 75	5-30 min	5-10	0.05-0.1%
15-minute	Workhorse	< 28 or > 72	15 min-2 hrs	3-6	0.1-0.2%
1-hour	Big catch	< 25 or > 75	1-8 hrs	0-2	0.2-0.5%

Session-aware thresholds (Melbourne time)

Session	Melbourne Time	Min Score	Mode
Asia	11am - 7pm	5/5	Observe only
London open	7pm - 12am	4/5	Alert
London-NY overlap	12am - 4am	3/5	PRIME ZONE
NY solo	4am - 9am	3/5	Active
Wind-down	9am - 11am	4/5	Review & adjust

Compounding roadmap

Phase	Period	Daily Target	Starting Capital	Expected End
Prove edge	Month 1	0.5%	$1,000	~$1,160
Scale target	Month 2-3	0.75%	$1,160	~$1,800
Full deploy	Month 4+	1.0%	$1,800+	Compounding

5. Mesh Resilience — Anduril Lattice Principle

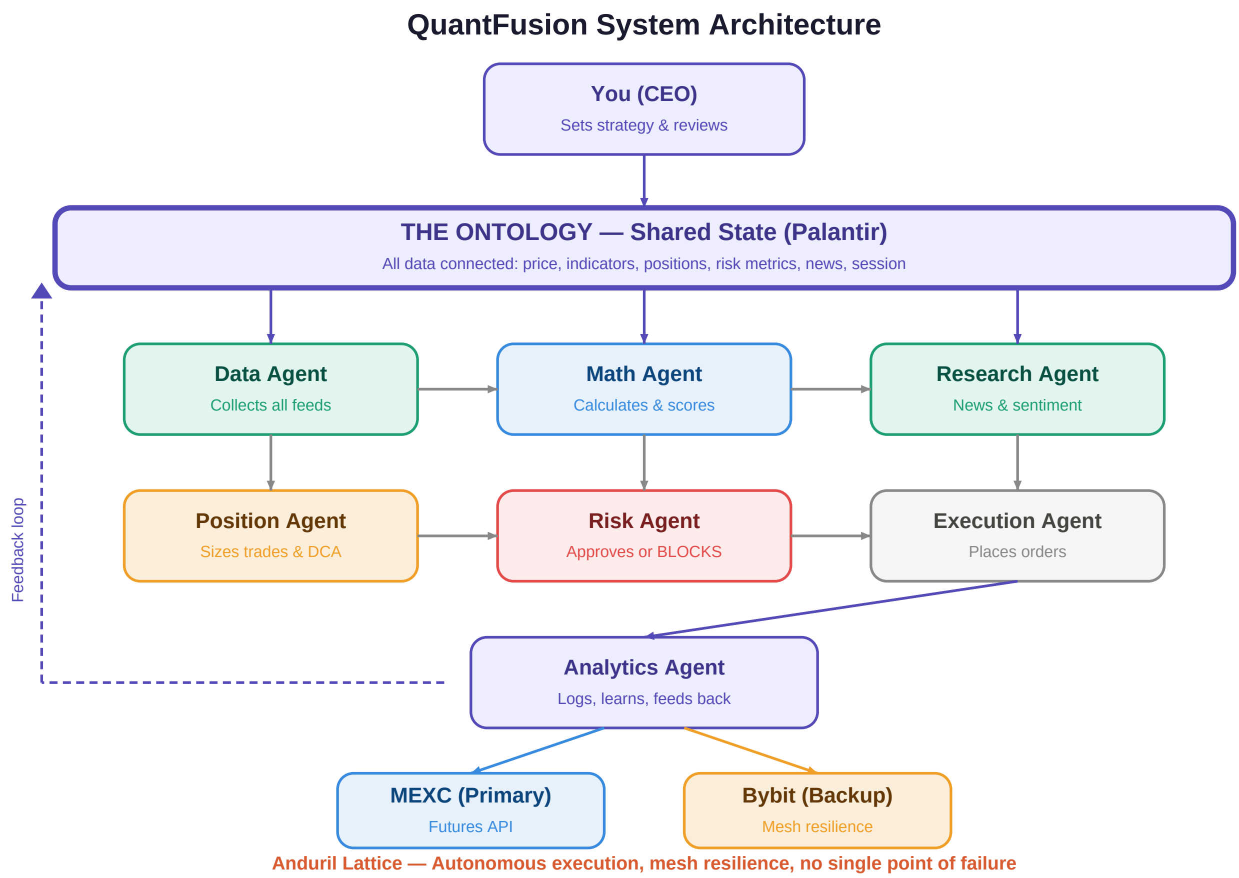
Every possible failure mode has a predefined fallback. The system degrades gracefully — it never crashes completely. Like Anduril's drone mesh, if one node goes down, others compensate.

Failure	Detection	Fallback	Recovery
MEXC API down	3 consecutive API errors	Switch to Bybit API	Auto-retry MEXC every 5 min
Price feed stale	No update > 10 seconds	Switch to backup feed	Alert CEO if > 2 min stale
News feed dies	No response > 15 min	Trade math-only, raise to 5/5	Resume normal when feed returns
Server crash	Heartbeat check fails	Exchange stops protect capital	Auto-restart on recovery
Internet drops	No connectivity	All positions have exchange stops	Resume when connection returns
Exchange maintenance	Scheduled or detected	Pause trading, hold positions	Resume when exchange is back
Abnormal spread	Spread > 3x normal	Pause entry, hold existing	Resume when spread normalises

The golden safety net: Every position ALWAYS has a stop loss set on the exchange itself. Even if every system, server, and connection fails simultaneously, the exchange-side stop loss protects your capital. An unprotected position never exists.

6. System Architecture Diagram

The complete QuantFusion system architecture showing how every component connects. Data flows from top to bottom. The Ontology sits at the centre connecting everything. The feedback loop from Analytics back to the Ontology enables continuous learning.



7. Key Principles

1. The system's primary job is saying NO.

Most signals get rejected. A system that trades all the time is a system that loses money. Quality over quantity — even with high-frequency, each trade must meet the checklist.

2. Position sizing is a formula, not a feeling.

Every position is calculated from risk budget and ATR. Never 'I feel like going bigger on this one.' Math decides size, always.

3. Same signal, different correct action.

RSI at 14 during a normal dip = buy. RSI at 14 during an exchange collapse = don't touch it. Context matters. That's why the Research Agent exists.

4. Compounding small wins is the business model.

Not home runs. Not 10x trades. Consistent 0.05-0.2% trades, 10 times a day, compounding daily. The casino model.

5. Higher timeframe always overrides lower.

If the 1-hour chart says bearish, the 5-minute chart cannot go long. The big picture sets the rules.

6. When the daily target is hit, stop.

Don't give back profits. Lock the win. Walk away. The system enforces this automatically.

7. Every failure has a fallback.

No single point of failure. Mesh resilience. Exchange-side stops. The system survives anything.

8. Start small, prove the edge, then scale.

Month 1: prove it works. Month 2-3: scale the target. Month 4+: scale the capital. Never skip this progression.

This is NOT financial advice. Trading involves risk. No system guarantees profits. Start with paper trading, validate statistically, then deploy with capital you can afford to lose.